Reward-Driven Decision Making

USING REINFORCEMENT LEARNING TO OPTIMIZE ROUTES

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ABOUT ME & THE PROBLEM

LinkedIn

Practical Experience & Applied Science Research

I work at **Aerton, Inc.** focused on Logistics & Field Service software for rural team.

We manage internal Fleet routing, dispatching, and tracking. Trying to reduce empty miles.

I'm also a PhD Student at MTSU - Computational Data Sciences.

My research is on Reinforcement Learning (RL) & Optimization models.

My research is Applied Sciences – Meaning finding solutions of current real-world problems.

The real-world environment is messy!

Our customers are trying to find repeatable operational ways to be better routing decisions.

In short, they strive to find an optimal solution to ever changing requirements.

Most businesses face these problems.

The difficulty is that the problems are a sequence of decisions not a single decision.

Solving the Problem is Sequential

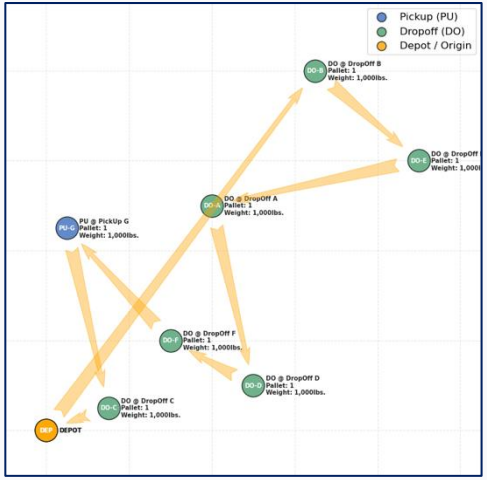
OPTIMIZATION IS EVERYWHERE

Why “Shortest Path” Often Fails

Real operations involve trade-offs, not just <u>distance!</u>

The closest next step is not always the best overall step.

Constraints and <u>downstream</u> consequences matter!



WHAT IS REINFORCEMENT LEARNING?

Learning Decisions from Experience

RL is a Subset of Machine Learning ML.

A model that allows a system to learn decisions from experience.

RL tries Actions and receives feedback in form of rewards.

It gradually improves its decision making over time.

It learns by interacting, not just being told the answer.

BOTTOM LINE

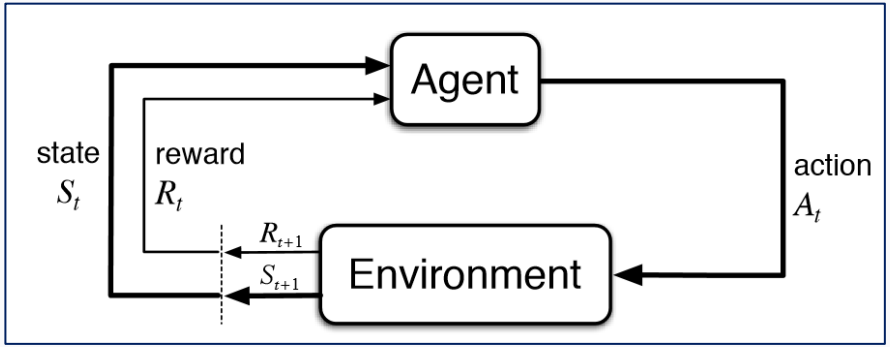
Learn by interacting, not just being told the answer.

Receive reward for good behavior!

THE CORE RL LOOP — WHAT IS THE AGENT

A Continuous Cycle of Improvement

1. **Observe** the current state
2. Choose an **Action**
3. Receive Feedback / **Reward**
4. Update behavior - Learn
5. Repeat



OBSERVE
↓
CHOOSE ACTION
↓
RECEIVE REWARD
↓
UPDATE
↓
REPEAT

RL QUESTIONS THAT MATTER

Defining System Boundaries in Reinforcement Learning

What can the system see?

Observation Space

What can the system do?

Action Space

How do we measure?

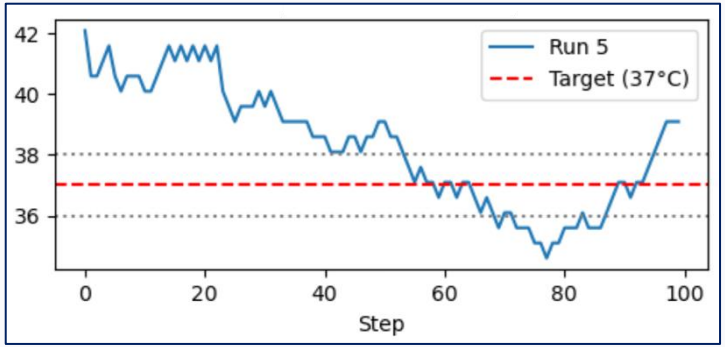
Reward

INTUITION: SIMPLE THERMOSTAT EXAMPLE

A Simple Example to Build Intuition on RL

PRIMARY GOAL

Keep the room near a target temperature while avoiding energy waste.



OBSERVATION

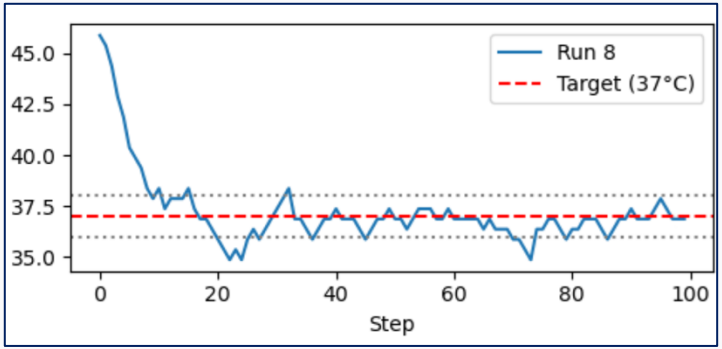
Current, target, and outside temperatures

ACTION

Increase | Decrease | Maintain Heat

REWARD

+ For target Proximity



SCALING COMPLEXITY – REAL WORLD APPS

What Changes When the Problem Gets Harder?

- More Information to Process – What is the State of Everything?
- More Possible Actions
- Stronger Long-Term Effects
- Increasing Number of Operational Constraints

SYSTEM SCALING

Better decisions become less obvious as the state space grows.

Especially for Humans!

ROUTE OPTIMIZATION IN AG IS HARD

Real-World Constraints in Rural Logistics

Pick-Up & Drop-Off Limitations!

We only have 20-26 Pallet Spots → Drop-Off First!

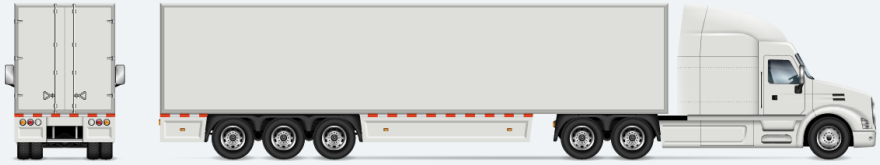
DOT Hours of Operation Constraints

Only allowed to drive 11hours

Fuel Costs Under Heavy Loads (*See Example*)

With heavy loads the fuel consumption goes up

Length	Width	Height	Volume	Max Cargo Weight	Pallets
48–53 ft	Up to 102.36 in	13–14 ft	Up to 3,489 cu ft	45,000 lbs	26



OPERATIONAL IMPACT

Cost, time, and service all matter simultaneously

Image Source: https://www.directliquidation.com

SEQUENCE QUALITY vs. DISTANCE / TIME

The best Route is Not Always the Shortes

A closer stop may create a **worse route later**!

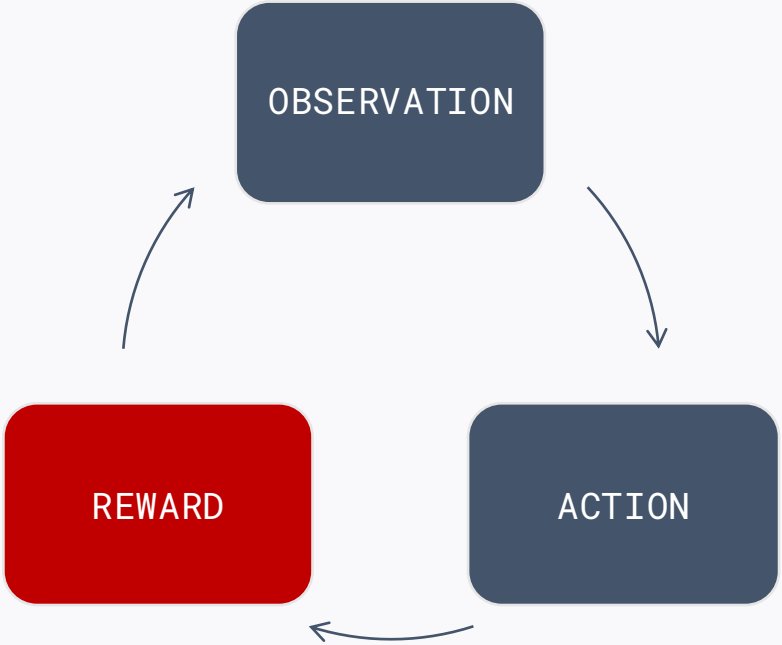
Pick-Up(s) increase Load / Weight
Drop-Off(s) reduce Load / Weight
→ **Sequence matters!**

Fuel and time **costs** change based on **sequence**.

MAPPING LOGISTICS TO REAL-WORLD

How Real-World Route Planning Becomes an RL Problem

The Secret is...



FEEDBACK

Reward-Driven Decision Making: Using Reinforcement Learning (RL) to Optimize Routes

REWARD SHOULD MATCH REAL GOAL

Aligning Feedback with Business Value!

Give Reward on Successful Drop-Off & Pick-Up:
This can be as simple as the count of Pallets / Weight or the actual revenue per load.

Penalize Time & Distance Costs:
This can is based on hourly rate of driver. Cost of Truck per Mile.

Penalize Fuel Consumption based on Weight:
Since we have the state of the truck we can estimate the fuel consumption based on the weight being hauled.

$$r_t = \underbrace{P(a_t)}_{\text{DO/PU Profit}} - \underbrace{C_{\text{drive}}\,\Delta time_t}_{\text{Drive Time Cost}} - \underbrace{C_{\text{gallon}}\,\frac{\Delta distance_t}{\text{MPG}(W_t)}}_{\text{Fuel Cost}}$$

BUSINESS ALIGNMENT

The reward function is the **Bridge** between business objectives and an RL-Model behavior

HOW IT LEARNS

The Iterative RL Training Process

$$r_t = \underbrace{P(a_t)}_{\text{DO/PU Profit}} - \underbrace{C_{\text{drive}}\,\Delta time_t}_{\text{Drive Time Cost}} - \underbrace{C_{\text{gallon}}\,\frac{\Delta distance_t}{\text{MPG}(W_t)}}_{\text{Fuel Cost}}$$

RL is un-supervised learning process!

Generate Route Scenarios – Random / Real
Let the model run an episode in a simulated environment

Measure the Resulting Rewards
The reward are used for internal policy updates and back propagation

Repeat across Many Scenarios and Episodes
In training we ran over 500 scenarios with 5,000 episodes each

TRAINING CYCLE

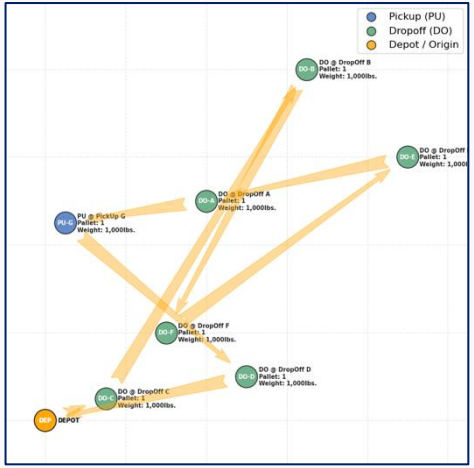
Learning is a continuous cycle of trial, feedback, and adjustment.

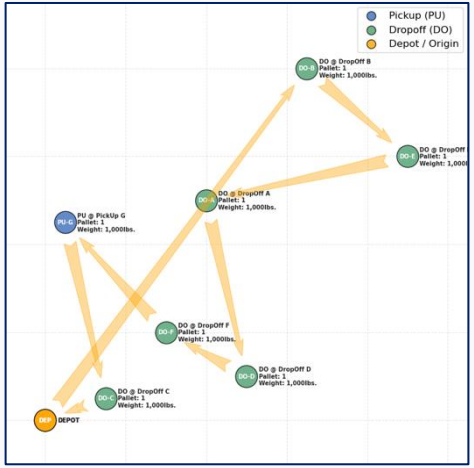
TRAINING PROGRESS EXAMPLE

Iterative RL Training Processes

Scenario

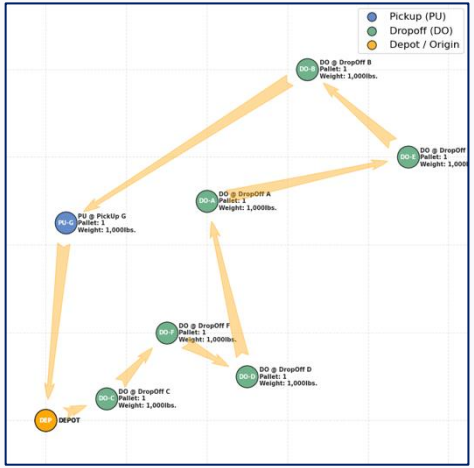
100 Episodes





250 Episodes

500 Episodes



EARLY TAKEAWAYS FROM RESEARCH

Lessons Learned From Applying RL to Logistics

Reward Design Is Everything
I noticed that the model would exploit any loophole or miss-configuration in the feedback-loop.

Simulation Fidelity Matters
If the environment isn't realistic; the model will not generalize to the field well.

RL will Find Hidden Shortcuts
During evaluation I noticed shortcuts that human heuristics have missed.

RESEARCH INSIGHT

The quality of the model is directly tied to the realism of its training environment.

MAIN TAKEAWAYS RL: COMPLEX ROUTING

The Future of Decision-Driven Operations Using RL

RL is About Sequential Decisions

It's the perfect tool for problems where one choice affects the next.

Reward Design Is the Bridge

It's how we translate business goals into a language the model understands.

Practical RL Requires Realism

Success in the field depends on the fidelity of the training environment.

QUESTIONS?

Thank you for your time!

FEEDBACK
Thank you!

PLEASE CONNECT
LinkedIn

SLIDES / CODE
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RL VS. SIMPLE HEURISTICS

The Value Proposition of RL over Traditional Rules

[vs] **Complex trade-offs** between distance, time, and load.

[vs] **Long-term consequences** of early route choices.

[vs] **Adaptive behavior** in changing environments.

VALUE PROPOSITION

RL finds solutions that simple rules often miss.

MAPPING LOGISTICS TO THE RL FRAMEWORK

How Route Planning Becomes an RL Problem

OBSERVATION

What the model knows about the route right now (locations, loads, (locations, loads, time).

ACTION

Which stop to visit next from the list of pending jobs.

REWARD

How good that choice was based on distance, time, and service service goals.

EPISODE

One full route from the depot to all stops and back to return.

OBSERVATION SPACE IN ROUTE OPTIMIZATION

Complex Inputs for Real-World Decisions

As the problem scales from a simple thermostat to complex logistics, the model requires a much richer set of data to make informed, global decisions.

ROUTE STATE VECTOR

Current location (GPS)

Candidate stops (Pending jobs)

Travel distance/time estimates

Completed vs. Uncompleted stops

ACTION SPACE: CHOOSING THE NEXT MOVE

Expanding the Set of Choices

[+] **Go to stop A, B, C...**: The model must evaluate every pending job.
pending job.

[+] **Return to depot**: A critical decision for route completion.

[+] The model is **repeatedly choosing** the next move.

DECISION CYCLE

The model is repeatedly choosing the next move from an expanding set of possibilities.

ENCODING BUSINESS RULES INTO THE PROBLEM

RL Learning Inside a Business Structure

[!] **Weight caps and pallet limits** as hard constraints for every route.
route.

[!] **Hours-of-service limits** for driver safety and regulatory compliance.
compliance.

[!] **Service priorities** and travel cost trade-offs.

STRATEGIC ALIGNMENT

RL is not replacing business logic; it is learning inside it.